

Capstone Project Overview

The document contains six use cases, each from a distinct industry domain, designed to demonstrate practical, medium-difficulty applications of core LangChain components.

Every use case is grounded in a realistic business problem and specifies which LangChain building blocks — LLMs, Prompt Templates, Chains, Agents & Tools, Memory, Document Loaders, Vector Stores, API Integration, and Evaluation/Debugging with LangSmith — are combined to deliver the solution.

Each entry below follows a consistent structure: a use case name, a problem description, the LangChain components involved, a short implementation workflow, and the expected business outcome.

Select **any one Use Case** as your capstone work for this training program.

Use Case Summary

S.No	Domain	Use Case Name
1	Banking & Financial Services (KYC/AML)	Regulatory Document Q&A Assistant for KYC Onboarding
2	Healthcare & Life Sciences	Clinical Literature Research Copilot
3	E-commerce & Retail	Conversational Product Recommendation Assistant
4	Legal Services	Contract Clause Extraction & Risk Flagging Agent
5	Human Resources & Recruitment	Resume Screening & Candidate Q&A System
6	Insurance	Claims Processing Assistant with Policy Lookup

1) Regulatory Document Q&A Assistant for KYC Onboarding

Description

An internal assistant that lets compliance analysts ask natural-language questions against a bank's KYC/AML policy manuals, RBI/FATF circulars, and customer onboarding SOPs, returning grounded answers with source citations instead of requiring manual document search.

Implementation Workflow

- Document Loaders ingest PDFs/DOCX of policy manuals and regulatory circulars.
- Text is chunked and embedded, then indexed in a Vector Store for semantic retrieval.
- A Prompt Template enforces a compliance-safe answer format (answer + cited clause + disclaimer).
- A Retrieval-Augmented Chain combines retrieved chunks with the LLM to generate grounded responses.
- LangSmith traces each query for accuracy review and hallucination monitoring by the compliance team.

Business Outcome

Analysts get instant, citation-backed answers to onboarding queries, cutting policy lookup time and reducing inconsistent interpretation of KYC rules across teams.

2) Clinical Literature Research Copilot

Description

A research assistant for clinical staff that searches and summarizes recent medical literature, trial results, and drug interaction data, combining internal knowledge bases with live web lookups for the most current findings.

Implementation Workflow

- Document Loaders pull in published trial PDFs and internal research notes.
- Content is embedded and stored in a Vector Store for fast semantic search across studies.
- An Agent equipped with a web-search Tool decides when to retrieve internally versus fetch newer external data.
- Conversation Memory lets researchers ask follow-up questions that build on prior results within a session.

Business Outcome

Researchers get faster, more comprehensive literature reviews with fewer manual searches, while keeping a session-aware thread across multi-step investigations.

3) Conversational Product Recommendation Assistant

Description

A shopping assistant embedded on a retail site that holds a multi-turn conversation to understand customer preferences and recommends products by querying the live product catalog API in real time.

Implementation Workflow

- Conversation Memory tracks stated preferences (budget, size, style) across the chat session.
- A Prompt Template structures the assistant's tone and constrains it to catalog-backed suggestions only.
- A Chain sequences preference-extraction, catalog querying, and response generation steps.
- API Integration connects the Chain to the store's live inventory and pricing API for accurate, in-stock results.

Business Outcome

Shoppers receive personalized, context-aware product suggestions that reflect real-time stock and pricing, improving conversion and reducing bounce from irrelevant results.

4) Contract Clause Extraction & Risk Flagging Agent

Description

An agent-driven workflow for legal teams that scans uploaded contracts, extracts key clauses (termination, liability, indemnity), and flags deviations from a firm's standard playbook for attorney review.

Implementation Workflow

- Document Loaders parse uploaded contract files (PDF/DOCX) into structured text.
- A Chain segments the contract and extracts named clause categories using targeted prompts.
- An Agent uses a comparison Tool to check extracted clauses against a standard clause playbook and flag risky deviations.
- LangSmith logs each extraction run, enabling legal ops to audit accuracy and refine prompts over time.

Business Outcome

Legal teams cut first-pass contract review time significantly and gain a consistent, auditable flagging process for non-standard terms.

5) Resume Screening & Candidate Q&A System

Description

A recruitment tool that indexes incoming resumes, allows recruiters to query candidates by skill or experience in natural language, and evaluates response quality against a scoring rubric.

Implementation Workflow

- Document Loaders ingest resumes in bulk from PDF and DOCX formats.
- Parsed candidate profiles are embedded and stored in a Vector Store for semantic skill-based search.
- A Chain handles query interpretation, candidate ranking, and summary generation for recruiters.
- LangSmith evaluation runs score the Chain's ranking outputs against a labeled test set to monitor drift and bias.

Business Outcome

Recruiters shortlist qualified candidates faster with natural-language search, while HR retains an evaluation trail to justify and audit screening decisions.

6) Claims Processing Assistant with Policy Lookup

Description

A claims-desk assistant that helps adjusters interpret policy coverage, check claim eligibility against policy terms, and pull real-time claim status from the insurer's core system via API.

Implementation Workflow

- API Integration connects the assistant to the policy administration and claims-status systems.
- An Agent selects the right Tool — policy lookup, eligibility checker, or claims-status API — based on the adjuster's question.
- A Chain composes the retrieved data into a clear, structured eligibility explanation.
- Memory retains claim context across a multi-question session so adjusters don't need to repeat claim IDs.

Business Outcome

Adjusters resolve coverage questions in a single conversation instead of switching between systems, shortening claims-cycle time and improving consistency of eligibility decisions.